Root-Cause Investigation: Falling Hospital Revenue

Management memo | Fictional Mumbai hospital group | Training case using simulated data

Decision status: Analysis framework and synthetic inputs are complete. This memo is intentionally marked preliminary: no real operational dataset was supplied, so no statistical finding or causal driver can honestly be claimed yet.

1. Problem statement

Across four hospitals, management has observed a decline in the headline performance measure over the latest two quarters. The investigation defines the headline metric as monthly net recognized revenue excluding GST. The objective is to identify where the decline is concentrated, distinguish structural deterioration from one-off dengue seasonality, and quantify the contribution of operational and commercial drivers.

2. Data and controls

The starter pack contains simulated OPD visits, admissions, laboratory tests and billing records totalling 500,000 rows, plus a patient-ID crosswalk. The pipeline maps alternate IDs to canonical patients, normalizes free-text diagnosis into broad clinical groups, reconciles gross = taxable + GST, and uses taxable amount as net revenue. Because gross amount is GST-inclusive, adding GST again would overstate revenue.

3. Hypotheses to test

H1: completed OPD volume declined. H2: Andheri is disproportionately affected. H3: self-pay realization weakened more than other payer segments. H4: service mix shifted away from higher-value services. H5: OPD-to-admission conversion deteriorated. H6: the August dengue surge is a temporary shock rather than the structural explanation. The accompanying hypothesis log specifies measures, tests and decision rules.

4. Analytical approach

The notebook produces monthly trends and breakdowns by hospital, service and payer segment; uses Welch t-tests for continuous measures and chi-square tests for categorical mix; and fits an adjusted regression with hospital, service and payer controls. The driver bridge reconciles segment-level changes to the observed total change. Weekly four-week rolling revenue and a threshold alert provide an early-warning candidate.

5. What management should expect

Do not interpret a statistically significant association as proof of cause. The strongest candidate driver must have (i) a plausible operational mechanism, (ii) timing that precedes or coincides with the decline, (iii) a dose-response across sites/segments, (iv) robustness to seasonality and mix controls, and (v) a credible comparison group or intervention test. If no intervention or adequate pre-period data exists, describe the conclusion as an evidence-backed association, not a proven causal effect.

Management actions and ownership

6. Proposed action plan (targets are provisional until baseline is validated)

Owner	Action	Provisional measurable target
Hospital COO + site heads	Weekly review of completed OPD volume, cancellations and service-line shortfall	Recover ≥50% of validated OPD shortfall within 8 weeks
Revenue cycle head	Audit net realization, discounts, denials and payer-wise collections	Reduce avoidable revenue leakage by 20% within 90 days
Clinical operations head	Review OPD-to-IPD conversion by service line and clinical appropriateness	Improve eligible-patient conversion by 2 percentage points in
Finance controller	Lock GST logic, refunds and credit-note reconciliation	100% monthly GST reconciliation; zero double-counted GST
Data/BI lead	Deploy weekly 4-week rolling revenue alert with site/service drill-down	Alert within 1 week of sustained ≥8% deterioration; investigat

7. Limitations and sign-off gates

The simulated dataset is designed to demonstrate common data problems, not to estimate actual hospital performance. The current six-month window is too short for robust annual seasonality modelling. The admission-to-OPD ratio is directional because the supplied event tables are not linked by encounter ID. Before any executive decision, replace the synthetic files with approved extracts, confirm definitions and completeness, validate the ID crosswalk, include refunds/credit notes and obtain Finance sign-off on revenue recognition.

Conclusion: This pack provides a complete reproducible investigation scaffold. The real root cause, ≥80% driver-tree coverage and any rejected hypotheses must be populated only after running it on the hospital's actual de-identified data.